STILLSTACK · DESIGN PRECIS

StillStack design precis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SSK-PRC-001	Version	0.10
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stillstack		
Source	docs/02-concept.md @ 4e01efb		

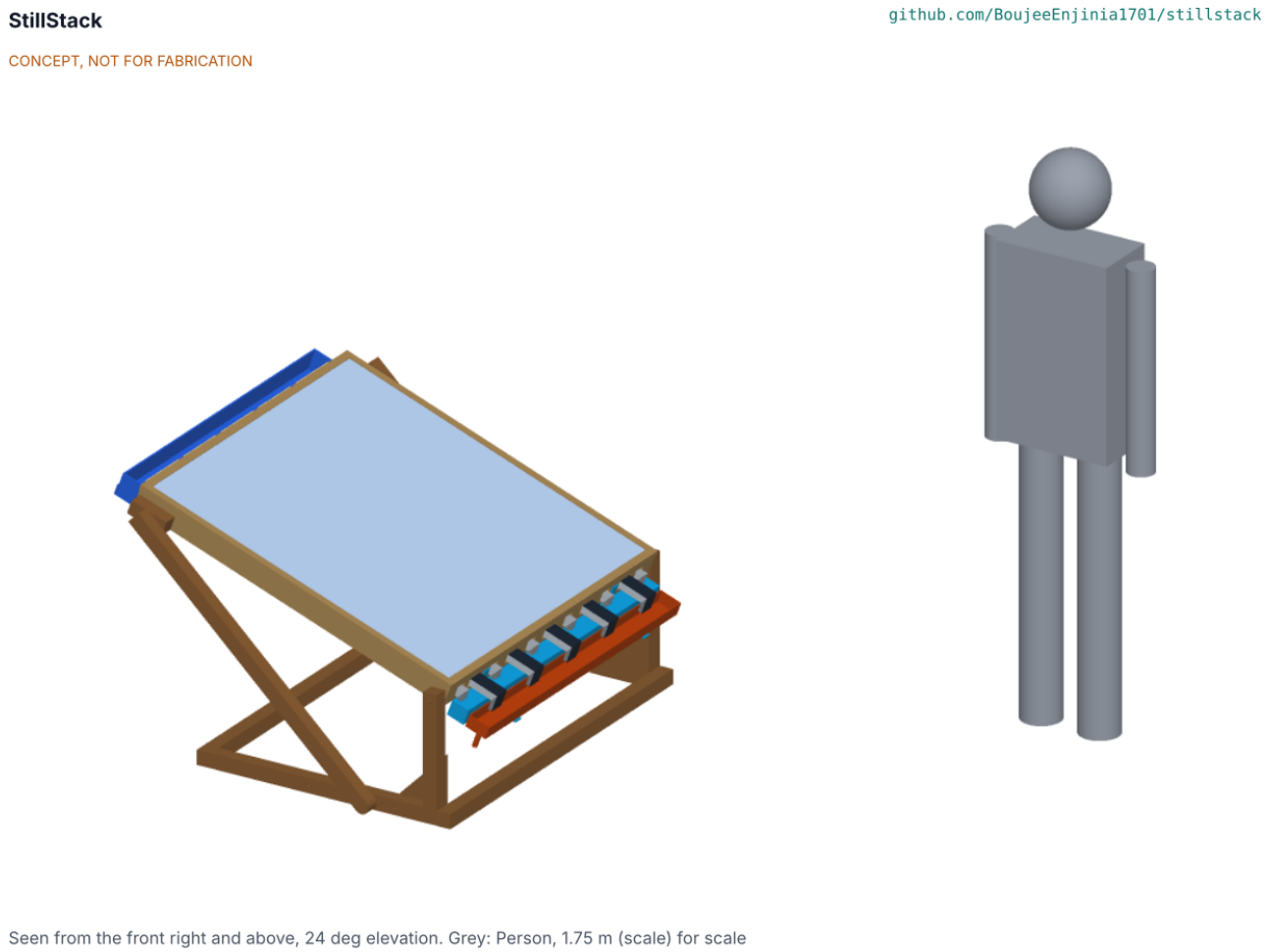
REVISION HISTORY

Version	Date	Author	Change
0.10	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: PR-002 restated as 1.2 or more (D-10); anchoring a hold point (D-09); cost USD 460.70 with the panel label (D-05)
0.9	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: stage 3 rails in PPS (SSK-DEC-001 D-01); this precis describes option A, the current design, while the concept selection between options A and B awaits Amish (SSK-OPT-001, D-03); mass 18.8 kg, cost USD 459.20
0.8	2026-10-02	Amish Chadha	Approved follow-ups carried into the design: mass 18.7 kg, cost USD 443.20, lip tabs and wick clips in the model
0.7	2026-10-02	Amish Chadha	Decisions of 2026-10-02: coating rule, wick fabric test and clips, PPS rails, high-edge lip, conductivity pass condition, partner, labels and rail colours
0.6	2026-10-01	Amish Chadha	

			Constructable design (SSK-DDR-003, open for Amish's review); budget as a value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update; decided design choices (SSK-DDR-001), spacer ribs, numbers from SSK-CAL-001, parametric model and GA drawing
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

StillStack design precis

StillStack is a tilted, glazed panel of 1 m2 containing four thin evaporation stages stacked like a sandwich: each stage is a wet cloth wick under a metal plate, a 6 mm vapor gap, and the next plate below. Sunlight heats the top plate, and the heat released when vapor condenses on each plate evaporates water from the wick under it, so the same solar energy is used four times. The TRL 3 calculation (SSK-CAL-001) gives about 10.3 L of distillate per m2 per day on a 5.5 kWh per m2 day, roughly two to three times a single-basin still, with a gained output ratio of 1.22 to 1.29 (PR-002, restated by Amish on 2026-10-25 as 1.2 or more for four stages, is met on paper). Dry stagnation reaches about 138 °C (142 °C in calm air), so stages 1 and 2 use materials rated to 150 °C and a stagnation cover is supplied (SSK-DDR-002). Value-engineering target: USD 355. Estimated cost of the constructable design: USD 460.70 (USD 105.70 over the target). This precis describes option A, the current design; a single-effect alternative (option B) is weighed in the concept options SSK-OPT-001, and the selection awaits Amish (SSK-DEC-001 D-03). How each part is made and fitted is in the prototype build plan (SSK-BLD-001); the changes that made the concept buildable are in SSK-DDR-003, open for Amish's review.

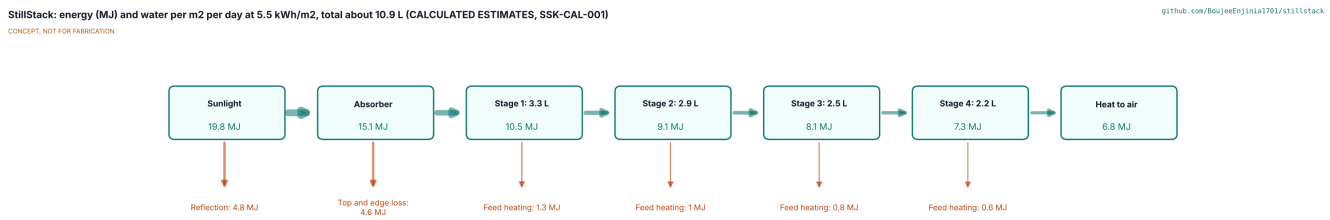


Seen from the front right and above, 24 deg elevation. Grey: Person, 1.75 m (scale) for scale

How it works

1. **Collect.** Sunlight passes through twin-wall polycarbonate glazing (solar transmittance about 0.80) and is absorbed by a black-coated aluminum plate. A 25 mm air gap under the glazing limits heat loss from the top.

2. **Evaporate, stage 1.** Cloth wick strips bonded to the underside of the absorber are kept wet with feed water. Heat conducts through the plate and evaporates water from the wick.
3. **Condense and reuse.** The vapor crosses a 6 mm gap and condenses on the top face of the next plate. The latent heat released there (about 2.35 MJ per kg at stage temperatures) conducts through that plate into the wick under it, which evaporates water into stage 2. The cascade repeats through four stages, each running about 4.5 K cooler than the one above at noon. Between 81 and 89 % of the heat crossing each gap at noon is latent.
4. **Hold the gaps.** Each gap has two side rails (PPS, long-term service 218 °C, in stages 1 to 3; printed polycarbonate in stage 4) and four silicone-cord ribs running down the slope at 194 mm pitch, so the 0.5 mm plates sag no more than about 0.5 mm. The wicks are cut into five strips with 10 mm dry breaks at every rib and rail.
5. **Reject heat.** The bottom plate is the last condenser. Folded aluminum fins on its shaded underside, painted matte black with the underside (SSK-DEC-001 D-07), reject the remaining heat (6.8 MJ per day, peak 286 W/m2) to air at about 25 K above ambient.
6. **Feed and drain.** Feed water drips from a user-filled container into a trough along the high edge. The wick tails pass out through openings in the high wall, rise behind a foam closure and drop into the trough, and the wicks are fed by capillary action and gravity down the slope. Excess salty water leaves the low end of each wick into a brine gutter that sits outboard of, and lower than, the distillate manifold.
7. **Collect distillate.** Condensate runs down each condenser plate to its zigzag low edge, which leads it to six tongues at the rib lines and corners. The tongues pass through notches in the low wall and turn down through slots in a lidded distillate manifold, which drains through a spout into a clean covered container. Brine leaves on separate tongues at the strip centres, carrying the wick tails over the closed lid into the brine gutter (SSK-DDR-003, proposed, awaiting Amish).



Main components

Table 1. Main components, numbered to match the exploded view and bom/bom.csv.

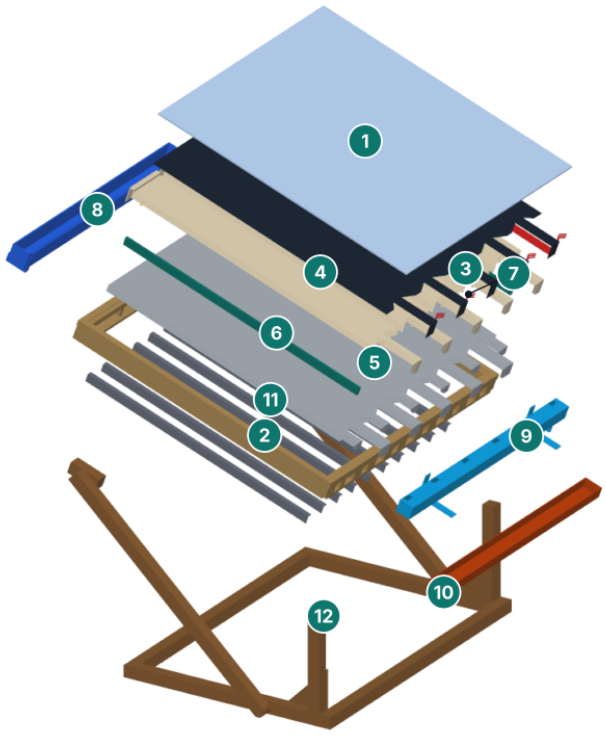
#	Component	Choice	Notes
1	Glazing	6 mm twin-wall UV-stabilized polycarbonate, 1.0 x 1.0 m	Inner sheet reaches about 109 °C at dry stagnation
2	Insulated frame	12 mm exterior plywood walls on a plywood base ring, with 25 mm foil-faced stone wool liner, 1,074 mm square outside; glazing held by an aluminium trim	Stone wool replaces PIR (SSK-DDR-002) because the absorber edge reaches about 144 °C at dry stagnation
3	Absorber plate	0.5 mm aluminum, high-temperature matte black paint on top	Stage 1 evaporator surface underneath
4	Wicks, 4 stages	Viscose-polyester or cotton nonwoven, about 1 mm, 1.0 x 1.25 m, cut into five strips	Needs a permeability of 3.0 x 10^-11 m2 or more and must tolerate 150 °C dry; product not yet selected
5	Condenser plates, 4	0.5 mm aluminum with a food-contact coating on the condensing face; 77 mm lip into the manifold	Coating rated to 150 °C on plates 1 and 2 and certified for drinking-water contact; the lowest-cost such coating (fluoropolymer cookware coatings first), else 316 stainless plates 1 and 2 (decided 2026-10-02); 4 mm downturned lip on each condensing plate's high edge (decided 2026-10-02)
6	Side spacer rails	12 x 7 mm, two per gap: stages 1 to 3 in PPS, long-term service 218 °C (stages 1 and 2 decided 2026-10-02, stage 3 added 2026-10-24 as SSK-DEC-001 D-01); stage 4 in printed polycarbonate	Split by SSK-DDR-002; ASA excluded by the stagnation temperature
7	Spacer ribs	7 mm food-grade silicone cord, four per gap at 194 mm pitch	Decided (SSK-DDR-002)
8	Feed trough	PVC or HDPE rain gutter section with end caps and drip valve	Fed from a user-filled container
9	Distillate manifold	Food-grade PP or HDPE channel with lid and silicone outlet tube	Separate outlet from the brine
10	Brine gutter	PVC gutter outboard of and below the distillate manifold, with drain hose	Brine to a soak-away or evaporation pond
11	Rear heat-rejection fins	Ten folded 0.5 mm aluminum strips, 30 mm deep	On the shaded underside of the bottom plate
12	Adjustable tilt stand	45 x 45 mm timber ground frame, front posts with a pivot bolt, fixed props pinned at one of six holes	10 to 35 deg in 5 deg steps (R11)
13	Sealant, fasteners and tubing	NSF/ANSI 51 or 61 silicone, stainless screws, silicone tube	Not shown in the model
14	Stagnation cover	Opaque white or aluminized tarpaulin 1.3 x 1.3 m with elastic edge cord	Fitted when the feed runs out and before maintenance; not shown in the model

StillStack: exploded view

github.com/BoujeeEnjinia1701/stillstack

CONCEPT, NOT FOR FABRICATION

1 Glazing, twin-wall polycarbonate
2 Insulated frame
3 Absorber plate, black
4 Wick strips, 4 stages
5 Condenser plates, 4
6 Side spacer rails
7 Spacer ribs, silicone cord
8 Feed trough
9 Distillate manifold
10 Brine gutter
11 Rear heat-rejection fins
12 Adjustable tilt stand



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Key numbers

All values come from SSK-CAL-001 (quasi-steady stack model integrated over a 5.5 kWh per m2 design day) and are calculated estimates, not measurements.

Table 2. Key numbers per m2 of aperture.

Quantity	Value	Requirement
Solar input	19.8 MJ per day	
Optical loss	4.8 MJ (τα = 0.76)	
Top loss	4.5 MJ	
Feed heating lost with brine	3.6 MJ	
Heat rejected at the bottom	6.8 MJ, peak 286 W/m2 (painted underside, D-07)	
Distillate by stage	3.33, 2.86, 2.50, 2.20 L (quasi-steady)	
Daily distillate	10.3 L after warm-up deduction (10.9 L quasi-steady); range 8.6 to 13.2 L (quasi-steady) for 4.5 to 6.5 kWh per m2	PR-001 (R1) at risk, 3 % margin
Gained output ratio	1.22 to 1.29	PR-002 (R2, 1.2 or more since 2026-10-25) met on paper
Plate temperatures at noon	73, 69, 64, 60 and 55 °C	
Dry stagnation, stage 1	about 138 °C (142 °C calm); stage 3 about 107 °C calm (PPS rails since D-01)	PR-008 (R8) at risk (150 °C and 110 °C ratings)
Feed and brine	about 27.2 L feed, 16.3 L brine per day; brine 58 to 67 g/L	PR-006 (R6) at risk (wick feed)
Panel mass	18.8 kg dry, 22.8 kg wet; stand 8.9 kg	PR-009 (R9) met
Parts cost	USD 460.70 against a USD 355 value-engineering target	PR-012 (R12) USD 105.70 over the target

Key design choices

These choices were decided by Amish on 2026-09-25 (SSK-DDR-001 and SSK-DDR-002).

- **Four stages with 6 mm gaps.** Decided. Three stages would give about 9.1 L per day; six stages about 14.9 L on paper.
- **Air-cooled fins under the bottom plate.** Decided. The bottom plate stays near 58 °C at noon, which is adequate for four stages.
- **Aluminum plates with a food-contact coating on the condensing face.** Decided. The coating product is not yet selected; see open questions.
- **Spacers in ASA or polycarbonate, not PETG.** Decided. SSK-CAL-001 shows that only polycarbonate is close to adequate, because stages 1 to 3 exceed the rating of ASA at dry stagnation.
- **Tilt of 20 deg on an adjustable stand.** Decided. The stand pivots at the low edge and sets the tilt with rear struts from 10 to 35 deg.
- **Aperture of 1 m2 and a value-engineering target of USD 355.** Decided. The budget was kept at $250 in SSK-DDR-001, raised to $320 in SSK-DDR-002 and topped up to $355 by Amish on 2026-09-26; on 2026-10-01 Amish set it as a value-engineering target, not a limit. The constructable design is estimated at USD 416.20.

- **Intermediate silicone-cord ribs.** Decided (SSK-DDR-002). The calculation shows that side rails alone cannot hold a 6 mm gap; silicone is rated well above the stagnation temperature and wets poorly.
- **Dry stagnation handled by materials and a cover.** Decided (SSK-DDR-002). Everything in stages 1 and 2 is rated to 150 °C or more, the frame liner is stone wool, and an opaque stagnation cover is supplied with a shade rule.

Safety

> **Safety:** Distillate is not certified drinking water. Test it before drinking (at least conductivity and E. coli), store it in a clean covered container, and disinfect it (for example with chlorine) before use. Each stage's distillate conductivity test is a pass condition at TRL 4 before any water is drunk (decided 2026-10-02). Stages run below pasteurization certainty, brine can wick or splash into distillate channels, and volatile contaminants in the feed can carry over with the vapor. Long-term drinking of distillate needs remineralization.
>
> **Safety:** Use only food-contact rated materials on every surface that touches distillate: coated or stainless plates, PP or HDPE channels, and NSF/ANSI 51 or 61 silicone. Never use lead-containing solder, treated timber or recycled containers of unknown history in the water path.
>
> **Safety:** The absorber reaches about 80 °C in normal use and about 140 °C when the wicks run dry. Do not touch the glazing or open the stack in sun. Fit the stagnation cover (BOM line 14) before maintenance and whenever the feed container runs empty, and never leave a dry panel uncovered in sun.
>
> **Safety:** Unanchored, the panel and stand can tip over at about 16.5 m/s of wind from behind (SSK-CAL-001 section 9). Staking the stand or ballasting it with 40 kg is a hold point before it is filled or faces the sun (D-09).

- Brine is strongly saline. Discharge it away from crops, wells and soil that must stay fresh.

Open questions

Decided by Amish on 2026-10-02 (SSK-DEC-001); the questions that remain are what the tests will show:

- Coating (R7): the lowest-cost coating certified for drinking-water contact (NSF/ANSI 61 or an equivalent food-contact statement) in hot salty water and rated to 150 °C dry; fluoropolymer (PTFE type) cookware coatings are checked first; if none meets both, 316 stainless for plates 1 and 2. Which product passes is still to be found.
- Wick fabric (R6): one polyester or polyester-viscose nonwoven and one glass-fibre fabric are shortlisted from datasheets and compared by a wicking-rise test; the cheaper one that reaches 3.0 x 10^-11 m2 and tolerates 150 °C dry is chosen. Wick fixing (R10): clips at the high and low ends, no adhesive.
- Rail polymer (R8, now PR-008): PPS for the stage 1 and 2 rails and stop blocks (2026-10-02), and since 2026-10-24 for stage 3 too, because printed polycarbonate softens at 113 °C and stage 3 reaches 110 °C in calm air (SSK-DEC-001 D-01, decided under Amish's standing instruction of 2026-10-23).
- Brine and feed separation (R4): the zigzag edge detail of SSK-DDR-003 (A1) and a 4 mm downturned lip on each condensing plate's high edge (A2); each stage's distillate conductivity at TRL 4 decides whether they work.
- Partner: a university water or desalination lab first; first candidate to approach, the Politecnico di Torino group behind the 2018 floating multistage still; a coastal NGO for field use once the distillate passes its tests. Nothing is agreed yet.
- Outlets and rails are marked DISTILLATE and BRINE, and the stage 1 and 2 rails have their own colour, so outlets are not mixed and a low-temperature rail is not fitted in a hot stage.

Design data: general arrangement SSK-DWG-002, sizing SSK-CAL-001, parametric model, blueprint sheet, interactive 3D model, cutaway.